C23-AIM-AI-CM-CAI-CCB-CCN–**102**

23008

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DCME-FIRST YEAR EXAMINATION

ENGINEERING MATHEMATICS—I

Time : 3 Hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

1. If $A = \{1,2,3\}$ and $f : A \to B$ is a function such that $f(x) = x^2$, then find range of f.

2. Resolve $\dfrac{x}{(\quad +1)(\quad +2)}$ into partial fractions.

3. If $A = \begin{bmatrix} 1 & 3 \\ & \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ & \end{bmatrix}$, then find $3A - 2B$.

4. Evaluate $\begin{vmatrix} \cos\theta & -\sin\theta \\ q & q \end{vmatrix}$

5. If $A = \dfrac{5}{\ }$ and $B = \dfrac{1}{\ }$, then show that $A + B = \dfrac{\pi}{4}$.

6. Prove that $\dfrac{1-\cos 2A}{\ } =$

7. Find the modulus of the complex number 3 + 4*i*.

8. Find the distance between the parallel lines $2x+3y+5=0$ and $2x+3y+4=0$.

9. Evaluate $\lim_{x\to 2}\frac{x^3-8}{x-2}$

10. Find the derivative of $3\cos x+\log x+5$ with respect to *x*.

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

11. Solve the following equations by using Cramer's rule :

$3x+2y+2z=5,\ 2x+y-z=1, x+y-z=0$

12. *(a)* Prove $\frac{\sin 4\theta+\sin 2\theta}{\cos 4\theta+\cos 2\theta}=\tan 3\theta$

(b) Prove that $\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)=\tan^{-1}\left(\frac{2}{9}\right)$

13. *(a)* Solve $\cos\theta+\sin\theta=\sqrt{2}$

(b) Solve the triangle *ABC* with $a=2, b=2\sqrt{3}, c=4$.

14. *(a)* Find the centre and radius of the circle $x^2+y^2+4x-6y-20=0$.

(b) Find the equation of the conic whose focus is (1,1) directrix is $x+y+1=0$ and eccentricity is $\frac{1}{2}$.

15. *(a)* Find the derivative of $2\sin x + 3\tan^{-1} x - 7\log x$ with respect to x.

(b) Find the derivative of $x^2 e^x$ w.r.t x.

16. *(a)* If $y = x^x$, then find $\frac{dy}{dx}$.

(b) If $y = ae^x + be^{-x}$, then prove that $\frac{d^2y}{dx^2} - y = 0$.

17. *(a)* Find the slopes of the tangent and normal to the curve $y = 2x^2 + 3x + 2$ at $(1,7)$.

(b) If $s(t) = 16t - t^2$, is a displacement of a particle, then find its velocity and acceleration at $t = 2$ sec.

18. *(a)* Find the maximum or minimum value of the function $f(x) = 2x^2 - 3x + 5$.

(b) If an error of 0·05% is made in the radius of a circle, then find the percentage error in measuring its area.

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